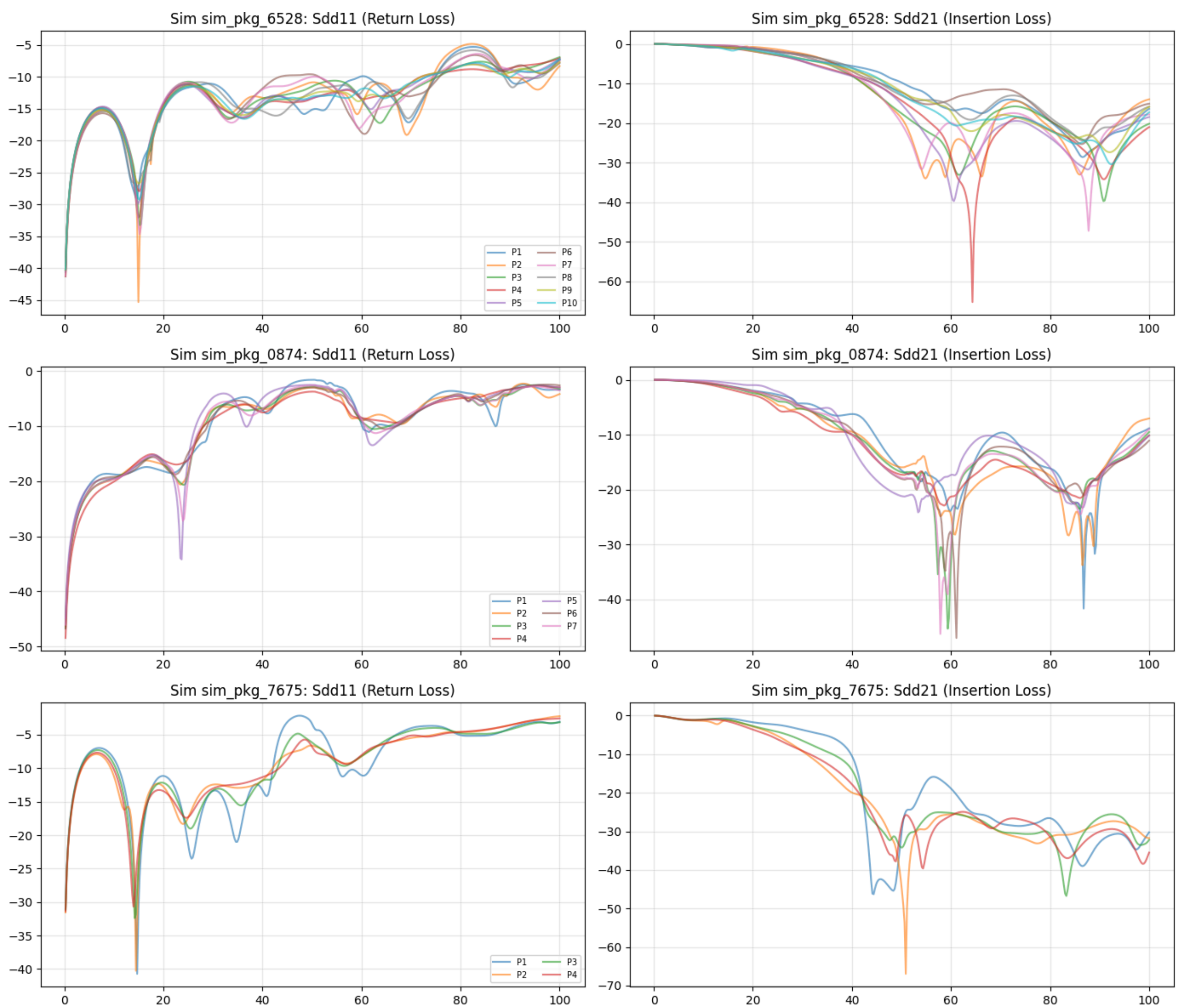
Progress till date

Changes in the data pipeline

In the previous data pipeline, I had extrapolation step because the script was using an interpolation from 0 to 100GHz, i changed it to 0.25 to 100GHJz hence eliminating the extrapolation step. and preserved the physical integrity of the S21 magnitudes as in TUHH dataset.

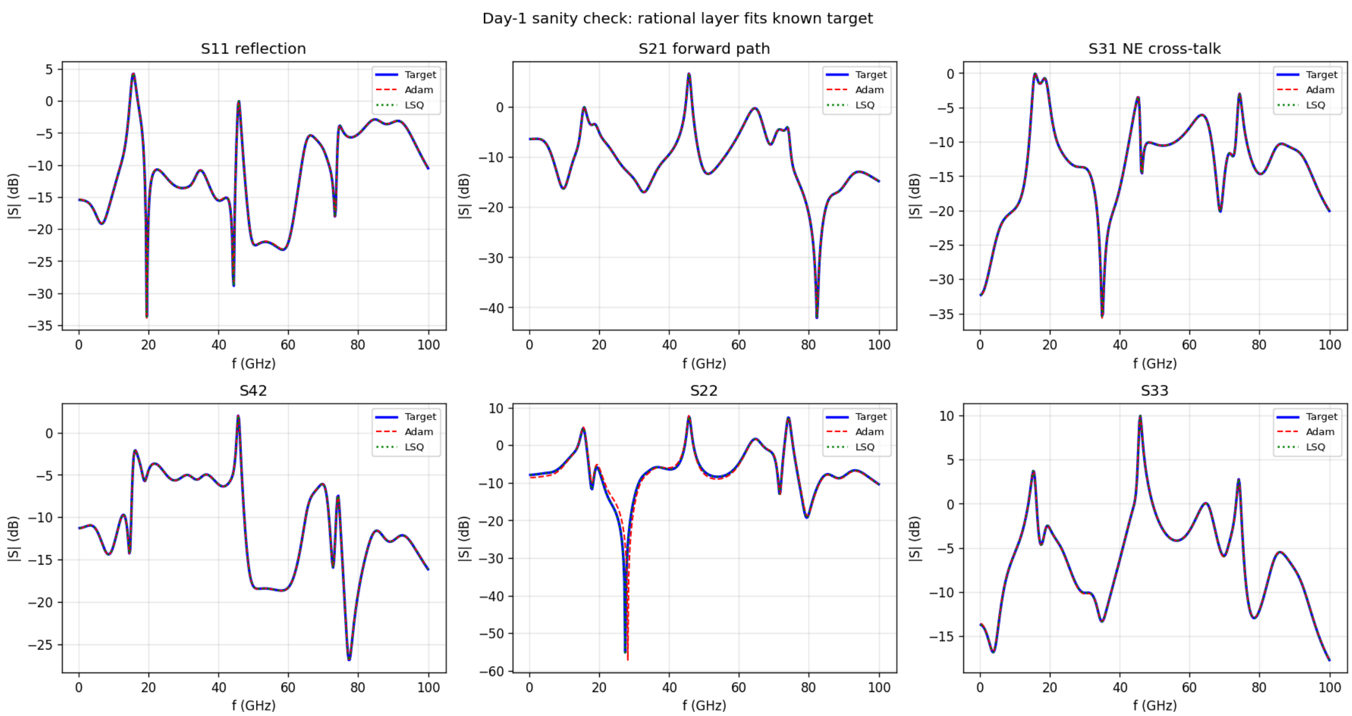
Instead of just clicing one 4x4 differetial matrix, I augmented the data by extracting the differential pairs for the whole simulation hence resulting in 6.9x datapoints than the previous pipeline.



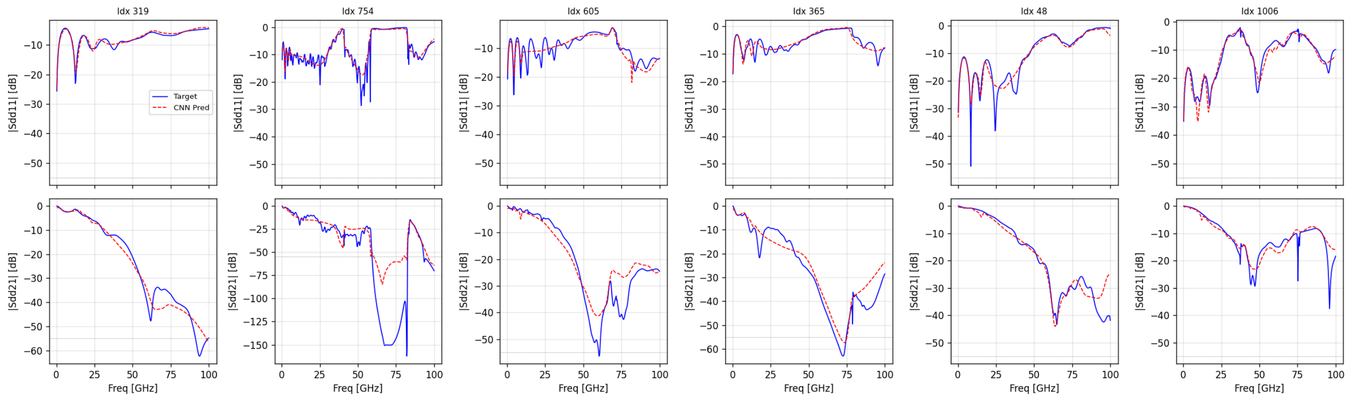
The above image is the samples comapring the extracted differential pairs with the raw ground truth dataset. But the complexity of the data was high (tried this while working on the forward model) where there are around 18 features with just 1916 sim in array and 1079 sim in Link - so the model converged well for the data pipeline but still it missed the sharp resonance dip in the region above 55GHz range where the noise is more and more data could actually solve this issue. But for now I just moved with the current forward model I tried.

Forward Model

Try 1: The foundational experiment was with the rational layer in the forward Neural Network design. I tried the advice you gave to create the synthetic dataset with poles and residues and try it against the rational net NN, it converges so well but also, the synthetic dataset was too simple,



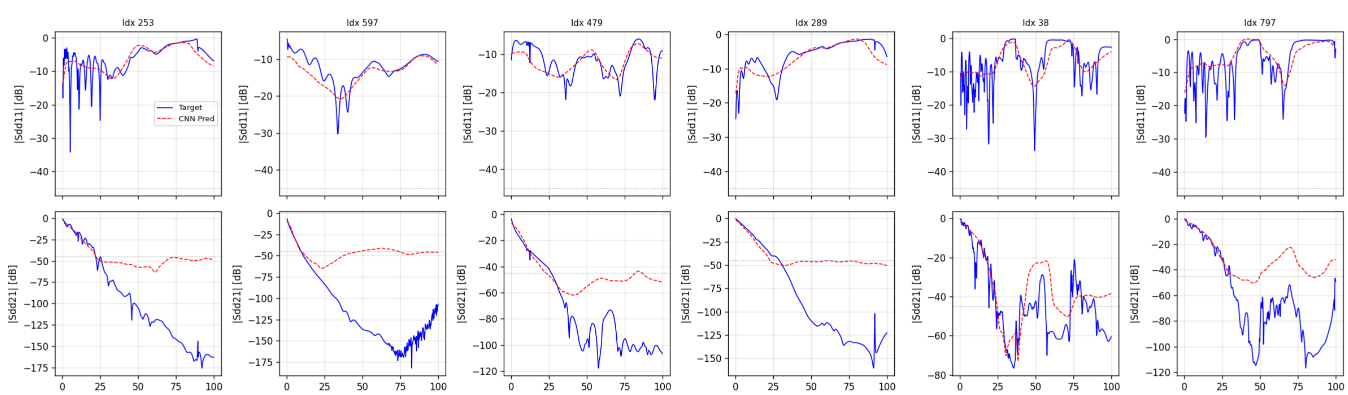
But when i tried this model with the TUIHH dataset it did not converge less than 3dB MAE. I later tried to use the 1D-CNN method and the direct sequence resnet actually performed better in array dataset with an average MAE of 2.12dB and it did track the resonance dip align the curves, again, it was not performing as expected in the range above 55GHz where the noise was dominating and the model struggled to converge. That is where the average MAE was taking a hit. I am still trying to make some improvements in parallel to the forward network while now working on the inverse model, but for now I am taking the best model I have at the moment and working on the inverse design. I tried the basic CVAE architecture in the inverse and constantly referring more papers to improve it to the project requirement.



As I said, I am still trying to improvise the models, I provide an update on the inverse model by next week as I will have the complete ablation study on the same.

Thew components I used in the forward 1D-CNN model for now are below,

- **Lorentz Reciprocity:** To preserve model capacity, structural reciprocity ($S_{ij} = S_{ji}$) was enforced by strictly predicting the 10 unique elements of the upper-triangular matrix.
- **Huber Loss Regression:** To prevent the optimizer from overfitting to HFSS numerical mesh noise at deep crosstalk nulls, the MSE loss was replaced by a Huber Loss on a decibel scale with a strict `-45.0 dB` clamp
- To ensure the neural network did not generate physically impossible (active) structures during inverse design, a differentiable Singular Value Decomposition (SVD) penalty was added to the training loss. This effectively bounded the predicted scattering matrices to $\sigma_{max} \leq 1.0$, mirroring SOTA Passivity Enforcement Layers (PEL)
  I will work on enhancing the link degradation because the average MAE for the kink dataset was around 2.5dB, but there is room for improvement. For now i am moving ahead with he array dataset model to check on how the inverse model performs, and I will work on the link dataset since the logic will be the same.



S21 on link was bad tbh but I will work on improvements.

Plan is to complete the inverse model by Mid of June - and I guess I am going in the right timeline as planned initially. i will provide you more update by end of next week.